taps, but I can also measure parameters that will influence electricity and water usage, such as the level of my rainwater storage tanks, and environmental parameters that impact power use such as room occupancy, temperature, and humidity. Figure 3.3 shows my most comprehensive sensor that allows me to measure a total of nine environmental parameters with one device.

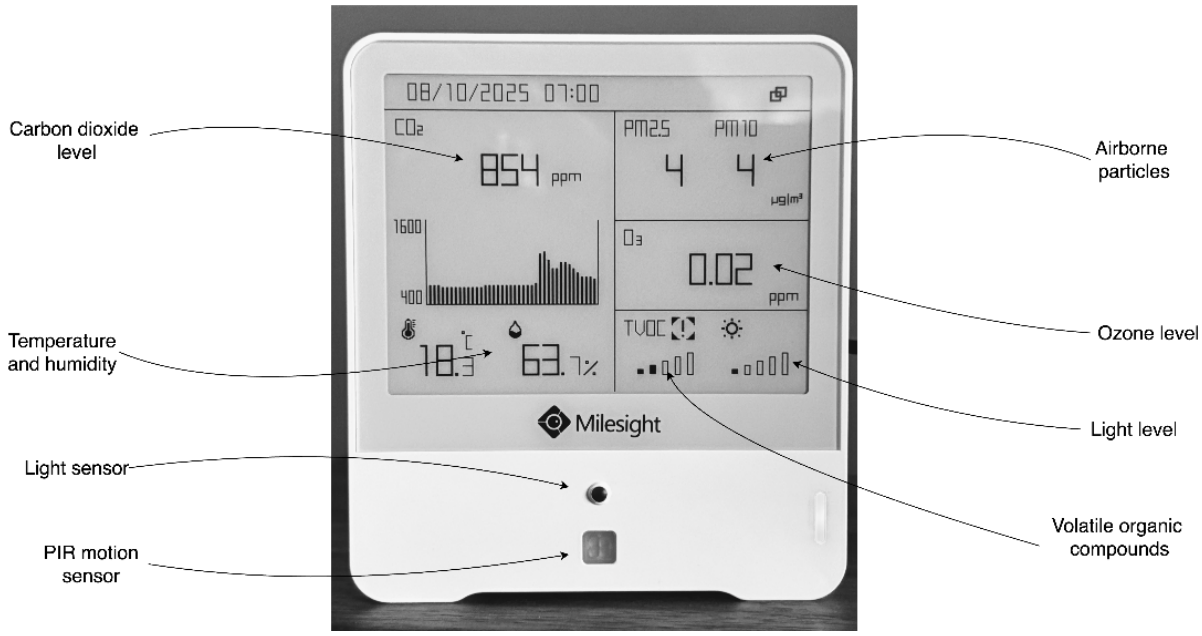


Figure 3.3 An environmental sensor that measures nine different parameters, displaying some on an E Ink display, as well as transmitting them over LoRaWAN.

Sometimes you will not be able to find the sensor that you need available to purchase off the shelf and will need to build your own. In appendix B, I show an example of how you can build an IoT sensor that measures temperature, humidity, and soil moisture using low cost electronic components.

3.3 Connecting sensors: communication technologies

Once you've established what types of sensors you require, you will need to get their data to your digital twin platform. Moving data between two points becomes surprisingly complex when multiplied across dozens or hundreds of sensors in diverse locations. The choice of communication technology significantly impacts your sensor selection, battery life, deployment cost, and